

C337/L337 – Introduction to Statistics with R/Biostatistics (3CR)

Indiana University South Bend

Course Syllabus

Instructor : Dr. Murli Nair	Lecture : Canvas
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Website : http://bioinformatics.iusb.edu	Office Hours: By appointment on zoom
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COURSE DESCRIPTION AND TENTATIVE SYLLABUS

The objective of this class is to introduce you to statistical methods used in data analysis. You will need to grasp the fundamentals of the R programming language and its environment. While we will touch upon advanced techniques, I won't delve deeply into them, as these are better suited for more advanced courses. The following outline provides a tentative schedule of the upcoming lectures/exams and may be subject to change.

Week	Topics	Chapters
Week0	Getting started and software installation	
Week1	Part 1: Introduction to Statistics	Chap 1 and 2
Week1	Part 2: R and RMarkdown	Notes provided
	Spring Break	
Week2	Part 1: Data reduction: Central Tendency/Variation	Chap 3,4,5
Week2	Part 2: Statistical Experimental Design	Notes provided
Week3	Part 1: Unit Exam1	
Week3	Part 2: Probability	Chap 6 and 20
Week4	Part 1: Sampling Distribution	Chap 7
Week4	Part 2: Parameter Estimation	Chap 8
Week5	Part1: Unit Exam2	
Week5	Hypothesis testing	Chap 9
Week6	Part 1: Correlation analysis	Chap 10
Week6	Part 2: Regression analysis	Chap 11
Week7	Analysis of variance	Chap 12
Week8	Final Exam	Cumulative

TEXT & LAB BOOKS

- **Statistical Analysis: An interdisciplinary introduction to univariate and multivariate methods by Kachigan (Required)**
- **OpenIntro Statistics** <https://www.openintro.org/>
- **Make use of free notes available on the R page** www.r-project.org

ONLINE RESOURCES

Assessments/HW: All assessments and homework will be posted on Canvas. Assessments in the form of short quizzes will be available for a specified time only. Please make sure to review the lectures and read the chapters well before attempting the quiz.

You are encouraged to visit the R home page at www.r-project.org.

For off-campus access, the same information will be accessible from both campus and home through IU's web-based Canvas system (<http://canvas.iu.edu>). As a student in my class, you are already registered as a Canvas user. However, you must activate your student computer account before obtaining the login credentials necessary to access it. Please activate your computer account as soon as possible. A notable feature of Canvas is its email notification system, which will alert you whenever new messages are sent to your Canvas account. To set this up, log in to Canvas, go to the "In Touch" folder, click on the "Canvas mail" section, and then select the "change preferences" option. Be sure to provide an email address that you check frequently. Note that this email address does not necessarily have to be associated with your free IUSB account. Whenever I post new items on the Canvas site (lecture notes, review sheets, readings, quiz reminders, etc.), I will send a message to the entire class to notify you.

GRADING

You will be graded on online quizzes, homework, assignments, unit exams, and the final exam.

Activity	Contribution	Late Penalty	Letter Grade	Range
Quizzes, HW and Assignments	50%	1% per hour	A+	100% to 97%
Unit Exam(s)	50%		A	<97% to 93%
Cumulative Final Exam			A-	<93% to 90%
Total	100%		B+	<90% to 87%
			B	<87% to 83%
			B-	<83% to 80%
			C+	<80% to 77%
			C	<77% to 73%
			C-	<73% to 70%
			D+	<70% to 67%
			D	<67% to 63%
			D-	<63% to 60%
			F	<60% to 0%

All questions regarding scores and/or grades need to be resolved within one week of the grade being posted.

ATTENDANCE

This is an online course, where you listen to the lectures and do the HW/assessments on your own. There are no points for attendance.

MAKE-UPS

Make-up opportunities will not be available beyond the scheduled time. In the event that you are unable to participate in a scheduled assessment, homework assignment, or exam due to a VALID excuse, your Final Exam grade will be considered in lieu of the corresponding quiz or assessment grade.

ACADEMIC INTEGRITY

As a student, you have the right to learn in an environment that fosters intellectual and ethical growth. However, this right also entails a corresponding responsibility for you to uphold ethical behavior. Consequently, your conduct must adhere to the Indiana University Code of Student Rights, Responsibilities, and Conduct (<http://www.iusb.edu/~judicial/>)."

Cheating, Plagiarism, Or Collusion Will Not Be Tolerated.

- While students are encouraged to engage in discussions about assignments with peers from both this class and others, it is imperative that the actual coding/assignment is completed individually.
- Sharing your code/assignment, even temporarily, with another student constitutes Collusion.
- Copying, whether in part or in whole, of the code/assignment (e.g., from a tutor, the internet, a student in the same or different class, or materials from a previous semester) amounts to Plagiarism.

Use of External Resources or Assistance: Any work you submit for grading must be your own, without any unauthorized assistance, which encompasses but is not limited to:

- Seeking help from other current or former students. Reading or sharing another student's work, even with the intent of mutual assistance, is prohibited. Sharing one's work with another student may be reported as "facilitating academic misconduct" (see the Code of Student Rights, Responsibilities, and Conduct at <https://studentcode.iu.edu/responsibilities/academic-misconduct.html>).
- Engaging the services of a tutor who is not currently affiliated with IU South Bend.
- Utilizing online platforms that offer solutions to questions or prompts which could be incorporated into your work. Examples of such platforms include chegg.com, stackoverflow.com, and ChatGPT.

Violation of the Ethical Code will result in a grade of "0" for the relevant assignment, and may lead to additional disciplinary measures, including receiving a failing grade for the entire course and the potential for dismissal from the University.

"Because dishonesty not only harms the individual but also impacts all students and undermines the University's integrity, policies regarding academic dishonesty will be rigorously enforced with zero tolerance."

All work should be your original product, unless explicitly noted otherwise. Any materials you reference or take from others should be properly cited. Cheating, plagiarism, or fabrication in any form will not be tolerated, regardless of any justification. For more detailed information see the [Student Responsibilities section of the Code of Student Rights, Responsibilities, and Conduct](#)[Links to an external site.](#). Academic misconduct will not be tolerated. The minimum consequence is failing the assignment. In a case of more serious offense, a student may fail the course. **Students should NOT present work from other courses in this class (i.e., using pieces of previous papers you have done is considered plagiarism).** I may use the services of Turnitin.com to check for originality of your written work.

Any work written, developed, created, or inspired by artificial intelligence (AI) is considered plagiarism and will not be tolerated. While the ever-changing developments with AI will find their place in our workforces and personal lives, in the realm of education and learning, this kind of technology does not belong. This is because the use of AI robs us of the opportunity to learn from our experiences and from each other, to play with our creative freedoms, to problem-solve, and to contribute our ideas in authentic ways. In a nutshell, college is a

place for learning, and this class is specifically a space for learning how to improve our writing. AI simply cannot do that learning for us. **You may be given an F in the course if you use AI.**

Cases of plagiarism will be dealt with as described in the Indiana University Academic Handbook and the Code of Student Rights, Responsibilities and Conduct. Plagiarism is any submission of work that is not your own without proper documentation of your sources; this includes, but is not limited to, submission of the work of another author as your own, submission of information obtained from the Internet without citation of your sources, and submission of work containing text or content/ideas from any other source without citation of those sources.

Citing a source is NOT a license to lift text or quote passages verbatim - all written work you submit must be in your own words. Please also note that using an **idea** that you did not originate without citation is plagiarism just as much as using text written by someone else is, and is in some ways a worse violation of intellectual property rights. If you are found to have plagiarized on an assignment, you will receive a grade of "F" for the course and the incident will be reported to the Vice Chancellor for Student Affairs for notation in your permanent academic record.

UNIT EXAMS AND FINAL EXAM

Since this is an online course, all the lectures and short assessments will be conducted online. However, unit exams and the final exam will be conducted using Honorlock. If you need accommodation for exams please contact me in advance and I shall try to do my best to accommodate your requests.

DISABILITIES AND OTHER ACCOMMODATIONS

DISABILITY: "Any student who feels that an accommodation may be needed based on the impact of a disability should contact Disabled Student Services at 520-4832 in office AL 120B of the Administration Building. Staff will work to coordinate reasonable accommodations for students with documented disabilities."

RELIGIOUS ACCOMMODATION: If any student will require academic accommodations for a religious observance, please provide me with a written request to consider a reasonable modification for that observance by the end of the second week of the course. Contact me after class, during my office hours, or by individual appointment to discuss the issue. If after discussion we reach no consensus, either party or both should seek the advice of the Dean, and if no consensus is reached, then the advice of the Vice Chancellor of Academic Affairs ("VCAA"). Either the instructor or the student may appeal the VCAA's decision to the Office of Affirmative Action within ten business days of the determination.

WITHDRAWAL

The timeline for withdrawing from the class can be found in the following URL.
<https://southbend.iu.edu/students/registrar/calendars/2026-spring.html>

COMMUNICATING WITH ME

Send e-mails directly to **mnair@iu.edu**. In the subject of your email please write “337 student”, I will use this to filter mails from students into a separate folder.